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ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 29-03-2026**

Time: 09.00AM TO 12.00PM **GTM-32** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below



Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

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MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. If a is a minimum value of $\sin^2\theta - \sin\theta + \frac{1}{2}$ and $b = \lim\limits_{x\to\infty}\left(\sqrt{(x+1)(x+2)} - x\right)$ then $|2a+b|$ is equal to

1) 3 **2)** -2 **3)** 4 **4)** 2

2. If $f(x) = x^3 + e^{x/2}$ and $g(x) = f^{-1}(x)$ then value of $g'(1)$ is

1) 1 **2)** 2 **3)** 3 **4)** e

3. The integral $\int\limits_{\pi/6}^{\pi/3} \sec^{2/3} x \operatorname{cosec}^{4/3} x \, dx$ is equal to

1) $3^{\frac{5}{3}}$ **2)** $3^{\frac{5}{6}} - 3^{\frac{2}{3}}$ **3)** $3^{\frac{4}{3}} - 3^{\frac{1}{3}}$ **4)** $3^{\frac{7}{6}} - 3^{\frac{5}{6}}$

4. A Number is chosen at random from the set {1,2,3,……..2000} Let P be the Probability that the chosen number is a multiple of 3 or a multiple of 7 then the value of 500P is

1) 211 **2)** 214 **3)** 210 **4)** 250

5. Let $S_k = \sum\limits_{r=1}^{k} \tan^{-1}\left(\frac{6^r}{2^{2r+1} + 3^{2r+1}}\right)$. Then $\lim\limits_{k\to\infty} S_k$ is equal to

1) $\tan^{-1}\left(\frac{3}{2}\right)$ **2)** $\frac{\pi}{2}$ **3)** $\cot^{-1}\left(\frac{3}{2}\right)$ **4)** $\tan^{-1}(3)$

6. The mean and variance of 5 observations of an experiment are 4 and 5.2 respectively. From these observations three are 1, 2, and 6, then the remaining will be

1) 2, 9 **2)** 5, 6 **3)** 4, 7 **4)** 3, 8

7. Let the locus of the points from which the tangents drawn to $y = x^2$ make an angle of 45° with each other is $16y^2 - 16x^2 + ky + 1 = 0$. Then k is equal to

1) 24 **2)** 34 **3)** 54 **4)** 64

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8. Consider points $A(3,4)$ and $B(7,13)$. If P be a point on the line $y=x$ such that $PA+PB$ is minimum, then coordinates of P are

1) $\left(\frac{12}{7},\frac{12}{7}\right)$ **2)** $\left(\frac{13}{7},\frac{13}{7}\right)$ **3)** $\left(\frac{31}{7},\frac{31}{7}\right)$ **4)** $(0,0)$

9. Let $\alpha=\cos\left(\frac{8\pi}{11}\right)+i\sin\left(\frac{8\pi}{11}\right)$

Statement-I: $\text{Re}\left(\alpha+\alpha^2+\alpha^3+\alpha^4+\alpha^5\right)=-\frac{1}{2}$

Statement-II: $\sum_{k=0}^{10}\alpha^k=0$

1) Statement I is true, Statement II is true, Statement II is correct explanation for Statement I

2) Statement I is true, Statement II is true, Statement II is not correct explanation for Statement I

3) Statement I is true, Statement II is false

4) Statement I is false, Statement II is true

10. If $x f(x)=3f^2(x)+2$ then $\int\frac{2x^2-12xf(x)+f(x)}{\left(6f(x)-x\right)\left(x^2-f(x)\right)^2}dx$ equals

1) $\frac{1}{x^2-f(x)}+C$ **2)** $\frac{1}{x^2+f(x)}+C$ **3)** $\frac{1}{x-f(x)}+C$ **4)** $\frac{1}{x+f(x)}+C$

11. The directrix of an ellipse with eccentricity $\frac{1}{\sqrt{2}}$ is $x+y=1$. If the corresponding focus is (1, **2)** then the length of the minor axis is

1) $\sqrt{2}$ **2)** 2 **3)** $2\sqrt{2}$ **4)** 4

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12. Tangents are drawn from the point $A(-2,1)$ to the circle $x^2+y^2-4x-6y+8=0$ touching it at the points P and Q. Then the equation of the circle circumscribing the ΔPAQ is

1) $x^2+y^2-4y+1=0$ **2)** $x^2+y^2-4y-1=0$

3) $x^2+y^2-4x-6y-7=0$ **4)** $x^2+y^2-4x-6y+8=0$

13. Match the following options:

Column-I		Column-II	
A)	$(x-1)(x-3)+k(x+2)(x-4)=0(k\in R)$ has real roots for $k\in$	p)	$(-5,-1)$
B)	The integer K for which the quadratic equation $(x-k)(x-10)+1=0$ has integral roots are	q)	12,8
C)	The equation $x\in\left(0,\frac{\pi}{2}\right)$, $\sec x+\operatorname{cosec} x=k$ has real root if $k\in$	r)	$(-\infty,\infty)$
D)	The equation $x^2+2(k-1)x+k+5=0$ has roots positive and distinct, if $k\in$	s)	$\left(2\sqrt{2},\infty\right)$

1) $A\to r, B\to q, C\to s, D\to p$ **2)** $A\to q, B\to r, C\to s, D\to p$

3) $A\to p, B\to r, C\to s, D\to q$ **4)** $A\to p, B\to s, C\to r, D\to q$

14. Let S= {1,2,3,----9} for K=1,2,……..5 let N_K be the number of subsets of S, each containing five elements out of which exactly K are odd. Then $N_1+N_2+N_3+N_4+N_5$ is equal to

1) 126 **2)** 125 **3)** 252 **4)** 210

15. The number of terms common to the two A.P.'s 3, 7, 11,…..407 and 3, 9, 16, 23, ……709 is

1) 14 **2)** 16 **3)** 18 **4)** 20

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16. Equation of plane which passes through the point of intersection of lines $\frac{x-1}{3}=\frac{y-2}{1}=\frac{z-3}{2}=\lambda$ and $\frac{x-3}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ and at greatest distance from the origin is…

1) $7x+2y+4z=54$ **2)** $4x+3y+5z=50$

3) $3x+4y+5z=49$ **4)** $x+y+z=12$

17. Range of $f(x)=\sin^{-1}\left(\sqrt{x^2+x+1}\right)$ is

1) $\left(0,\frac{\pi}{2}\right]$ **2)** $\left(0,\frac{\pi}{3}\right]$ **3)** $\left[\frac{\pi}{3},\frac{\pi}{2}\right]$ **4)** $\left[\frac{\pi}{6},\frac{\pi}{3}\right]$

18. The minimum number of elements that must be added to the relation $R=\{(1,2)(2,3)\}$ on the set of natural numbers. So that it is an equivalence relation is …

1) 6 **2)** 7 **3)** 8 **4)** 9

19. The coefficient of the term independent of x in the expansion of $\left(1+x+2x^3\right)\left(\frac{3}{2}x^2-\frac{1}{3x}\right)^9$ is

1) $\frac{1}{3}$ **2)** $\frac{19}{54}$ **3)** $\frac{17}{54}$ **4)** $\frac{1}{4}$

20. A man walks a distance of 3 units from the origin towards the North East (N45°E) direction. From there he walks a distance of 4 units towards the North West (N45°W) direction to a point P. Then the position of P in the argand plane is

1) $e^{\frac{\pi}{4}+4i}$ **2)** $(3-4i)e^{i\frac{\pi}{4}}$ **3)** $(4+3i)e^{i\frac{\pi}{4}}$ **4)** $(3+4i)e^{i\frac{\pi}{4}}$

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SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. Let C be a curve passing through $M(2,2)$ such that the slope of the tangent at any point to the curve is reciprocal of the ordinate of the point. If the area bounded by curve C and line $x=2$ is A then the value of $\frac{3A}{2}$ is

22. Let $y=y(x)$ be solution of the following differential equation -

$e^y\frac{dy}{dx}-2e^y\sin x+\sin x\cos^2 x=0 \quad y\left(\frac{\pi}{2}\right)=0$. If $y(0)=\log_e\left(\alpha+\beta e^{-2}\right)$ then $4(\alpha+\beta)$ is equal to

23. Words of length 10 are formed using the letters A,B,C,D,E,F,G,H,I,J Let x be the number of such words where no letter is repeated; and let y be the number of such words where exactly one letter is repeated twice and no other letter is repeated then $\frac{y}{9x}$ is equal to

24. If $\alpha,\beta\neq 0$ and $f(n)=\alpha^n+\beta^n$ and $\begin{vmatrix} 3 & 1+f(1) & 1+f(2) \\ 1+f(1) & 1+f(2) & 1+f(3) \\ 1+f(2) & 1+f(3) & 1+f(4) \end{vmatrix}$

$=k(1-\alpha)^2(1-\beta)^2(\alpha-\beta)^2$ then k is equal to

25. From a lot of 10 items which include 3 defective items, a sample of 5 items is drawn at random. Let the random variable X denote the number of defective items in the sample. If the variance of X is σ^2 then $96\sigma^2$ is equal to

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PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

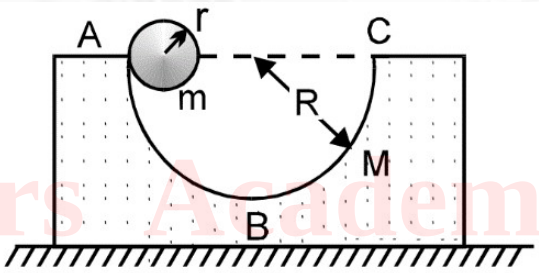
This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. The upper portion of an inclined plane of inclination α is smooth and the lower portion is rough. A particle slides down from rest from the top and just comes to rest at the foot. If the ratio of the smooth length to rough length is m : n, the coefficient of friction is

1) $\left[\frac{m+n}{n}\right]\tan\alpha$ **2)** $\left(\frac{m+n}{n}\right)\cot\alpha$ **3)** $\left(\frac{m-n}{n}\right)\cot\alpha$ **4)** $\frac{1}{2}$

27. A block of mass M with a semicircular track of radius R rests on a horizontal frictionless surface. A uniform cylinder of radius r and mass m is released from rest from the top point A. the cylinder slips on the semicircular frictionless track. The distance travelled by the block when the cylinder reaches the point B is



1) $\frac{M(R-r)}{M+m}$ **2)** $\frac{m(R-r)}{M+m}$ **3)** $\frac{(M+m)R}{M}$ **4)** None

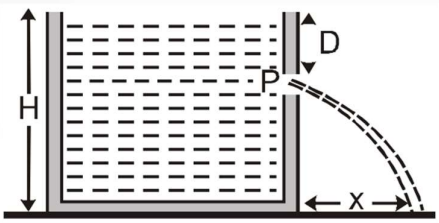
28. Two point masses m_1 and m_2 are connected at the ends of a light rigid rod of length L. The moment of inertia of the system about an axis through their centre of mass and perpendicular to the rod is

1) $\frac{1}{2}\left(\frac{m_1m_2}{m_1+m_2}\right)L^2$ **2)** $\left(\frac{m_1m_2}{m_1+m_2}\right)L^2$

3) $(m_1+m_2)L^2$ **4)** $\left[m_1^2+m_2^2\right]\left(\frac{m_1+m_2}{m_1m_2}\right)L^2$

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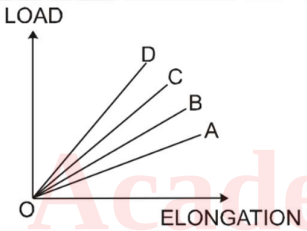
29. A tank is filled with water up to height H. Water is allowed to come out of a hole P in one of the walls at a depth D below the surface of water as shown in the figure. Express the horizontal distance x in terms of H and D



1) $x=\sqrt{D(H-D)}$ **2)** $x=\sqrt{\frac{D(H-D)}{2}}$

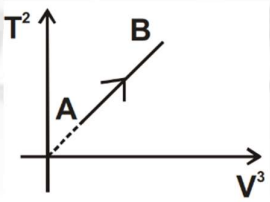
3) $x=2\sqrt{D(H-D)}$ **4)** $x=4\sqrt{D(H-D)}$

30. The load versus elongation graphs for four wires having same length and made by the same material is shown in the figure. The thinnest wire is represented by the line:



1) OC **2)** OD **3)** OA **4)** OB

31. If ideal diatomic gas follows the process, as shown in graph, where T is temperature in kelvin and V is volume $\left(m^3\right)$ then molar heat capacity for this process will be [in terms of gas constant R]



1) $\frac{7R}{2}$ **2)** $5R$ **3)** $\frac{19R}{6}$ **4)** $\frac{11R}{2}$

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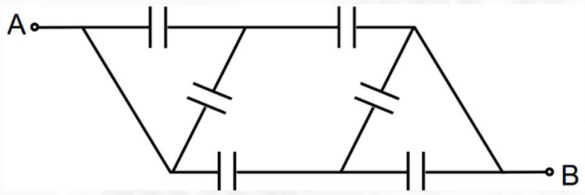
32. A mono–atomic ideal gas is compressed from volume V to V/2 through various process. For which of the following processes final pressure will be maximum

1) isobaric **2)** isothermal **3)** adiabatic **4)** PV^2 = constant

33. Two point charges placed at a distance r in air exert a force F on each other. The value of distance R at which they experience force 4F when placed in a medium of dielectric constant K = 16 is.

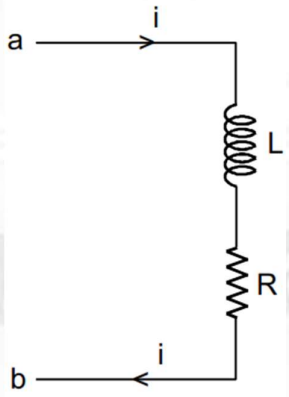
1) r **2)** r/4 **3)** r/8 **4)** 2r

34. A network of six identical capacitors, each of capacitance C is formed as shown below. The equivalent capacitance between the points A and B is



1) 3C **2)** 6C **3)** 3C/2 **4)** 4C/3

35. When the current in the portion of the circuit shown in the figure is 2A and increasing at the rate of 1 A/s, the measured potential difference $V_a - V_b = 8V$. However when the current is 2A and decreasing at the rate of 1 A/s, the measured potential difference $V_a - V_b = 4V$. The values of R and L are



1) 3 ohm and 2 Henry respectively **2)** 2 ohm and 3 Henry respectively

3) 10 ohm and 6 Henry respectively **4)** 6 ohm and 1 Henry respectively

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36. In a photoelectric experiment, with light of wavelength λ, the fastest electron has speed v. If the exciting wavelength is changed to $\frac{3\lambda}{4}$, the speed of the fastest emitted electron will become

1) $v\sqrt{\frac{3}{4}}$ **2)** $v\sqrt{\frac{4}{3}}$ **3)** less than $v\sqrt{\frac{3}{4}}$ **4)** greater than $v\sqrt{\frac{4}{3}}$

37. If $\lambda_{\min}$ is minimum wavelength produced in x-ray tube and $\lambda_{k\alpha}$ is the wavelength of k_α line. As the operating tube voltage is increased.

1) $(\lambda_{k\alpha} - \lambda_{\min})$ increases **2)** $(\lambda_{k\alpha} - \lambda_{\min})$ decreases

3) $\lambda_{k\alpha}$ increases **4)** $\lambda_{k\alpha}$ decreases

38. An astronomical telescope has an objective lens of focal length 100 cm and for relax eye adjustment magnification is 10. The separation between objective and eye piece lens is

1) 110 cm **2)** 90 cm **3)** 10 cm **4)** 125 cm

39. Two identical equi convex lenses made of the same material of refractive index 1.5 have the focal length 12 cm. These lenses are kept in contact and immersed in a liquid of refractive index 1.35. The combination behaves as

1) convex lens of focal length 27 cm **2)** concave lens of focal length 6 cm

3) convex lens of focal length 9 cm **4)** convex lens of focal length 6 cm

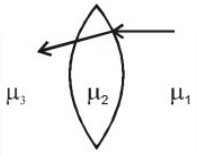
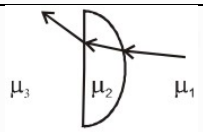
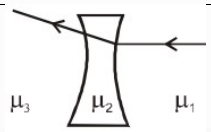
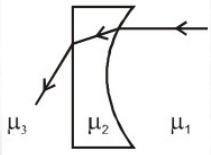
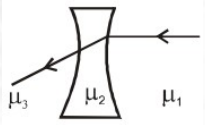
40. The velocity of water waves may depend on their wavelength λ, the density of water ρ and the acceleration due to gravity g. The method of dimensions gives the relation between these quantities as [where k is a dimensionless constant]

1) $v^2 = k\lambda^{-1}g^{-1}\rho^{-1}$ **2)** $v^2 = k\,g\,\lambda$

3) $v^2 = k\,g\,\lambda\,\rho$ **4)** $v^2 = k\,\lambda^2 g^{-1}\rho^{-1}$

41. Two transparent media of refractive indices μ_1 and μ_3 have a solid lens shaped transparent material of refractive index μ_2 between them as shown in the figures in Column II. A ray traversing these media is also shown in the figures. In Column I different relationships between μ_1, μ_2 and μ_3 are given. Match them to the ray diagrams shown in Column II.

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Column I		Column II	
(A)	$\mu_1 < \mu_2$	(p)	
(B)	$\mu_1 > \mu_2$	(q)	
(C)	$\mu_2 = \mu_3$	(r)	
(D)	$\mu_2 > \mu_3$	(s)	
		(t)	

1) (A) - p,r ; (B) - q,s,t ; (C) - p,r,t ; (D) - q, s
2) (A) – r,s ; (B) - r,s,t ; (C) - s,t ; (D) - p, q
3) (A) – r,s ; (B) - s,t ; (C) - r,s,t ; (D) - p, q
4) (A) - p,q ; (B) – p,q,r ; (C) - q,r,s ; (D) - r,s

42. Assertion: The nuclear density of nuclides ${}^{10}_{5}B, {}^{6}_{3}Li, {}^{56}_{26}Fe, {}^{20}_{10}Ne$ and ${}^{209}_{83}Bi$ can be arranged as $\rho^N_{Bi} > \rho^N_{Fe} > \rho^N_{Ne} > \rho^N_{B} > \rho^N_{Li}$

Reason: The radius of the nucleus is related to its mass number as $R = R_0 A^{1/3}$, where R_0 is a constant.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) A is not correct but R correct
2) Both A and R correct and R is the correct explanation of A
3) Both A and R are correct but R is not the correct explanation of A
4) A is correct but R is not correct.

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JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

43. Assertion: Alloys such as constantan and manganin are used in making standard resistance coils.
Reason: Constantan and manganin have very small values of temperature coefficient of resistance.
In the light of the above statements, choose the most appropriate answer from the options given below:
1) A is not correct but R correct
2) Both A and R correct and R is the correct explanation of A
3) Both A and R are correct but R is not the correct explanation of A
4) A is correct but R is not correct.

44. Statement1: Diffusion current in a p-n junction is greater than the drift current in magnitude if the junction is forward biased
Statement 2: Diffusion current in a p-n junction is from the n-side to the p-side if the junction is forward-biased.
In the light of the above statements, choose the most appropriate answer from the options given below:
1) Statement1 is not correct but Statement 2 correct
2) Both Statement1 and Statement 2 correct and Statement 2 is the correct explanation of Statement1
3) Both Statement1 and Statement 2 are correct but Statement 2 is not the correct explanation of Statement1
4) Statement1 is correct but Statement 2 is not correct.

45. Statement 1: A pendulum clock when taken to Mount Everest becomes fast.
Statement 2: The value of g (acceleration due to gravity) is less at Mount Everest than its value on the surface of Earth.
In the light of the above statements, choose the most appropriate answer from the options given below:
1) Statement1 is not correct but Statement 2 correct
2) Both Statement1 and Statement 2 correct and Statement 2 is the correct explanation of Statement1
3) Both Statement1 and Statement 2 are correct but Statement 2 is not the correct explanation of Statement1
4) Statement1 is correct but Statement 2 is not correct.

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SECTION-II
(NUMERICAL VALUE TYPE

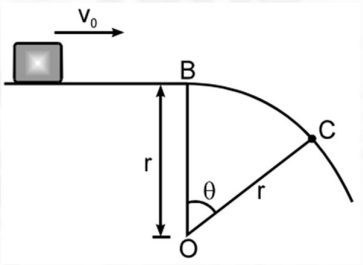
This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. Two particles A and B are situated 10m apart along X axis, B being farther right of A, at $t = 0$. Particle A is moving at 0.75 m/s parallel to +ve Y axis while B at 1 m/s along -ve X-axis. After a time t they come closest to each other. t is $n \times 10^{-1}$ sec. Find n.

47. A small frictionless block slides with velocity $0.5\sqrt{gr}$ on the horizontal surface as shown in the figure. The block leaves the surface at point C. The angle θ in the figure is $\cos^{-1}(3/x)$.

Find x



48. A ball of mass = 100gm is released from a height $h_1 = 2.5m$ from the ground level and then rebounds to a height $h_2 = 0.625m$. The time of contact of the ball and the ground is $\Delta t = 0.01\text{sec}$ The impulsive (impact) force offered by the ball on the ground is …. N (take $g = 9.8\,m/\text{sec}^2$)

49. A whistle whose air column is open at both ends has a fundamental frequency 500 Hz. The whistle is dipped in water such that half of it remains out of water. What will be the fundamental frequency now (in Hz)? (Speed of sound in air is 340 ms^{-1})

50. Two bodies are in equilibrium when suspended in water from the arms of a balance. The mass of one body is 36 g and its density is 9 g/cc. If the mass of the other is 48 g, its density in g/cc is (Assume arms lengths are equal)

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CHEMISTRY

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. The correct statements regarding the geometrical isomers of 2-butene is

(A) Cis-2-butene and trans-2-butene are not interconvertible at room temperature

(B) Cis-2-butene has less dipole moment than trans-2-butene

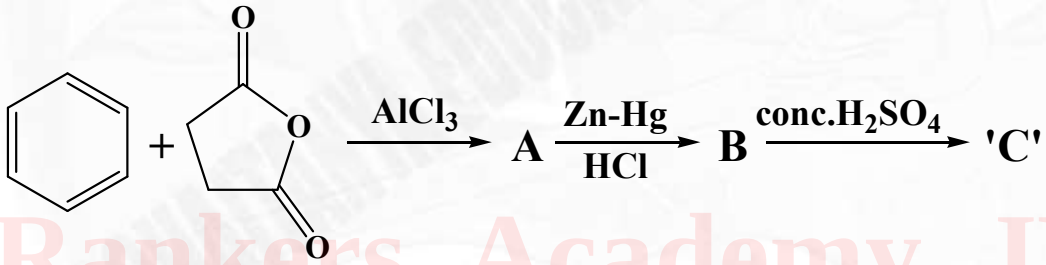
(C) Trans-2-butene is more stable than Cis-2-butene

(D) Cis-2-butene and trans-2-butene are stereo isomers.

1) A, B, C, D are correct **2)** only B is correct

3) A, C, D are correct **4)** A, B, C are correct

52.



Which of the following is correct about A, B, & C.

(A) A has 3-hetero atoms (B) B has 2-hetero atoms

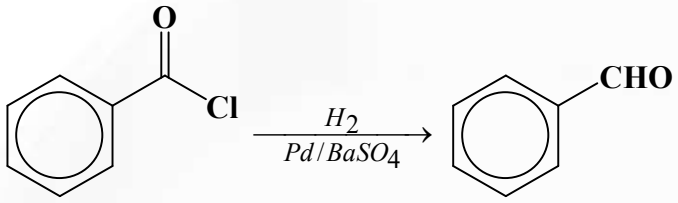
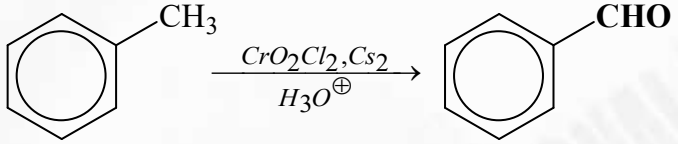
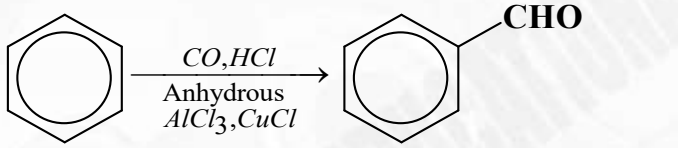
(C) C has 1-hetero atom (D) C has 7-Sp^2 hybrid orbitals

1) All are correct **2)** ABC are correct

3) ABD are correct **4)** BCD are correct

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JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
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53. Match List-I with List-II

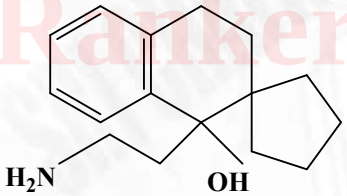
List-I		List-II	
A)	$RCN \xrightarrow[H_3O^{\oplus}]{SnCl_2,HCl} RCHO$	i)	Etard reaction
B)		ii)	Gattermann Koch reaction
C)		iii)	Rosenmund reaction
D)		iv)	Stephen reaction

Choose the correct answer from the options given below

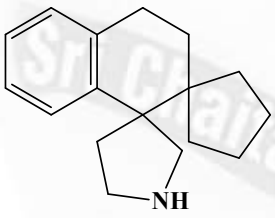
1) A-iv, B-iii, C-i, D-ii **2)** A-iii, B-iv, C-ii, D-i

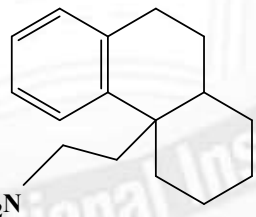
3) A-i, B-iii, C-ii, D-iv **4)** A-iii, B-iv, C-i, D-ii

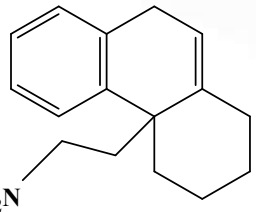
54. In The reaction given below

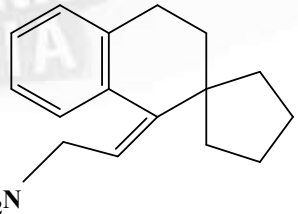


$\xrightarrow[(ii)KOH]{(i)HCl}$ $'B'$ (Major) ; $'B'$ is

1) 

2) 

3) 

4) 

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IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
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55. An organic compound [A], molecular formula $C_{10}H_{20}O_2$ was hydrolyzed with dilute sulphuric acid to give a carboxylic acid [B] and an alcohol [C]. Oxidation of [C] with $CrO_3 - H_2SO_4$ produced [B], which of the following structure are not possible for [A]?

1) $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-OCOCH_2-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_3$

2) $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-COO-CH_2-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_3$

3) $CH_3CH_2CH_2COOCH_2CH_2CH_2CH_3$

4) $(CH_3)_3-C-COOCH_2C(CH_3)_3$

56. Given below are two statements.

Statement-I: Fructose does not contain an aldehydic group but still reduces Tollen's reagent.

Statement-II: In the presence of base, fructose undergoes rearrangement to give glucose.

In the light of the above statements, choose the correct answer from the options given below.

1) Statement-I is false but Statement-II is true

2) Both Statement-I and Statement-II are true

3) Both Statement-I and Statement-II are false

4) Statement-I is true but Statement-II is false

57. $CaC_2 \xrightarrow{H_2O} 'A' \xrightarrow[\text{Fe-tube}]{\text{Red hot}} 'B' \xrightarrow[h\upsilon]{CH_3CH_2CH_2Cl} 'C' \xrightarrow[ii)H_3O^{\oplus}]{i)O_2} 'D' + 'E'(\text{ketone})$

Which of the following is correct

(A) 'A' reacts with Tollen's reagent to form red precipitate

(B) 'B' is also formed from phenol by dry distillation in the presence of Zn

(C) Ethanol is million times acidic than compound 'D'

(D) 'E' reacts with Grignard reagent followed by hydrolysis to form 3°-alcohol

1) B & D **2)** A & B **3)** C & D **4)** A, B, C & D

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IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
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58. On passing a gas 'X' through Nessler's reagent, a brown precipitate is obtained the gas 'X' is forms a complex with a cation in Leclanché cell. The correct statements among below are

A) The coordination number of metal ion in the complex is 4

B) The number of unpaired e^- in central metal in the complex is one.

C) The hybridization of central metal in the complex is sp^3

D) The complex is square planar

1) A, B, C **2)** A, C **3)** B, C **4)** A, C, D

59. The number of correct statements are

A) When di dentate ligand ethane -1, 2-diamine (en) is added to $\left[Ni\left(H_2O\right)_6\right]^{3+}$ complex in 3 : 1 ratio, the colour of complex is pale blue

B) The crystal field splitting Δ_0 depends upon the field produced by the ligand and charge on the metal ion.

C) The complex $\left[Co\left(NH_3\right)_5 SO_4\right]Br$ exhibit ionization isomerism.

D) The primary valency of central metal ion is non ionizable, but the secondary valency of central metal ion is ionizable.

1) B, D **2)** A, B, C **3)** B, C **4)** A, C

60. Which of the following is correctly represented

1) The angular momentum of the electron $mvr = \frac{nh}{\pi}$

2) The orbital angular momentum of '3p' orbital is $\frac{h}{\sqrt{2}\pi}$

3) The number of radial nodes for '4d' orbital is 3.

4) The radius of nth orbit $r_n = \frac{0.529 \times z^2}{n^2} A^0$

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THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
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61. Given below are two statements.

Statement-I: The single N – N bond is weaker than the single P – P bond

Statement-II: Phosphorous and Arsenic can form $d\pi - d\pi$ bond with highly electro negative elements [F, O, N]

In the light of the above statements, choose the correct answer from the options given below.

1) Statement-I is false but Statement-II is true

2) Both Statement-I and Statement-II are true

3) Both Statement-I and Statement-II are false

4) Statement-I is true but Statement-II is false

62. Match List-1 with List-II

List-I Symbol of elements		**List-II General outer electronic configuration of group**	
(A)	Og	1	ns^2np^5
(B)	Ts	2	ns^2np^3
(C)	Lv	3	ns^2np^6
(D)	Mc	4	ns^2np^4

Choose the correct answer from the following

1) A-3, B-1, C-4, D-2 **2)** A-2, B-4, C-1, D-3

3) A-1, B-2, C-3, D-4 **4)** A-4, B-3, C-2, D-1

63. A molecule with the formula $M\,x_2y_2$ has all its elements from p-block. Element M is mono atomic gas and its I.E. is equal to I.E. of Molecular Oxygen. x is lowest electron gain enthalpy in chalcogens y is strongest oxidizing agent in halogens. The shape and number of lone pairs on central atom are

1) Sea saw, 1 L.P. **2)** Square pyramidal, 2 L.P.

3) Square planar, 1 L.P. **4)** Tetrahedral, 1 L.P.

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THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
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64. Following are some statements about equilibrium process. The number of correct statements are

A) $H_2O(\ell) \rightleftharpoons H_2O(g)$ equilibrium process P_{H_2O} is constant at given temperature

B) $H_2O(s) \rightleftharpoons H_2O(\ell)$ equilibrium process melting point is fixed at constant temperature

C) Equilibrium is possible in both closed system and open system.

D) All measurable properties of the system remains constant.

1) A, B only **2)** B, C, D only **3)** A, B, D only **4)** A, C, D only

65. Predict which of the following entropy increases.

1) A liquid crystalizes into solid

2) Temperature of a crystalline solid is raised from 0K to 115K

3) $2H(g) \rightarrow H_2(g)$

4) $PCl_3(g) + Cl_2(g) \rightarrow PCl_5(g)$

66. Some statements about $K_2Cr_2O_7$ & $KMnO_4$. Choose the incorrect statement

1) Sodium dichromate is more soluble than potassium dichromate

2) Both $K_2Cr_2O_7$ & $KMnO_4$ oxidizes I^- into I_2 in acidic medium

3) $Cr_2O_7^{2-}$ ion is orange colour and MnO_4^{2-} is green colour is due to d-d transition

4) Both CrO_4^{2-} and MnO_4^- are in Tetrahedral structure.

67. Lanthanoid ions with $4f^6$ electronic configuration are

A) Sm^{2+} B) Eu^{3+} C) Pm^{2+} D) Gd^{3+} E) Nd^{2+}

1) A, B only **2)** B, C only **3)** A, C, D only **4)** B, C, D only

68. Identify the homoleptic complexes with diamagnetic substances

A) $\left[Fe(CN)_6\right]^{3-}$ B) $\left[Ni(CO)_4\right]$ C) $\left[Fe(CN)_5 NO\right]^{2-}$ D) $\left[Co(H_2O)_3 F_3\right]$

E) $[FeF_6]^{3-}$ F) $\left[Cr(H_2O)_6\right]^{3+}$

Choose the correct answer from the option given below

1) A, B, E **2)** B, C, F **3)** B only **4)** A, B, E, F

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JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
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69. Assertion: H_2O has high boiling point than H_2S

Reason: H_2O contain intramolecular H-bond, where as H_2S does not contain intramolecular H-bond

In light of the above statements, choose the most appropriate answer from the options given below:

1) A is not correct but R correct

2) Both A and R correct and R is the correct explanation of A

3) Both A and R are correct but R is not the correct explanation of A

4) A is correct but R is not correct.

70. Which of the following binary mixture shows positive deviation from Raoult's Law?

1) $C_6H_6 + C_6H_5CH_3$ **2)** $C_2H_5Br + C_2H_5Cl$

3) $C_2H_5OH + CH_3COCH_3$ **4)** $CHCl_3 + CH_3COCH_3$

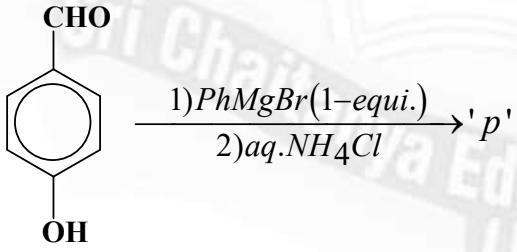
SECTION-II
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This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. 'X' gm of ethylamine is subjected to reaction with $NaNO_2 / HCl$ followed by water, evolved dinitrogen gas which occupied 2.24 L volume at STP. 'X' is $\ldots \times 10^{-1}\, gm$

72. The product of the following reaction is 'p'



The number of hydroxyl groups present in the product 'p' is

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JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
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73. The molar conductivity of 0.025 M methanoic acid is 46.1 S cm^2 $mole^{-1}$. Its degree of dissociation is $x \times 10^{-2}$. Where x is…

[Given $\lambda^\circ\left(H^+\right) = 349.6\, S\, cm^2 mole^{-1}$ and $\lambda^\circ\left(HCO\bar{O}\right) = 54.6\, S\, cm^2 mole^{-1}$

74. A first order reaction is found to have rate constant $k = 5.5 \times 10^{-14}$ sec^{-1} the half life of the reaction is $x \times 10^{11}$ sec, where x is ….

75. The density of 3 M solution of NaCl is 1.25 gr/ml. Calculate the molality of solution (nearest integer).

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42 RANK MD ANAS HT. No. 252046210*	45 RANK RAMIT GOYAL HT. No. 257001113*	52 RANK MAULIK JAIN HT. No. 252079407*	54 RANK GARV HT. No. 252056188*	59 RANK LARISSA HT. No. 252079071*	60 RANK ARYAN BALABADRULA HT. No. 256132077
63 RANK SAMYAJYOTI BISWAS HT. No. 255058456*	64 RANK AARUSH ANAND HT. No. 251006176*	72 RANK RUSHMITH BANDARI HT. No. 256168048	78 RANK KORIKANA RASAGNYA HT. No. 256057046	87 RANK LAKSHYA SHARMA HT. No. 252070079*	91 RANK AVANEESH BANSAL HT. No. 251113130*	95 RANK KAVYA AGGARWAL HT. No. 252079121*

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